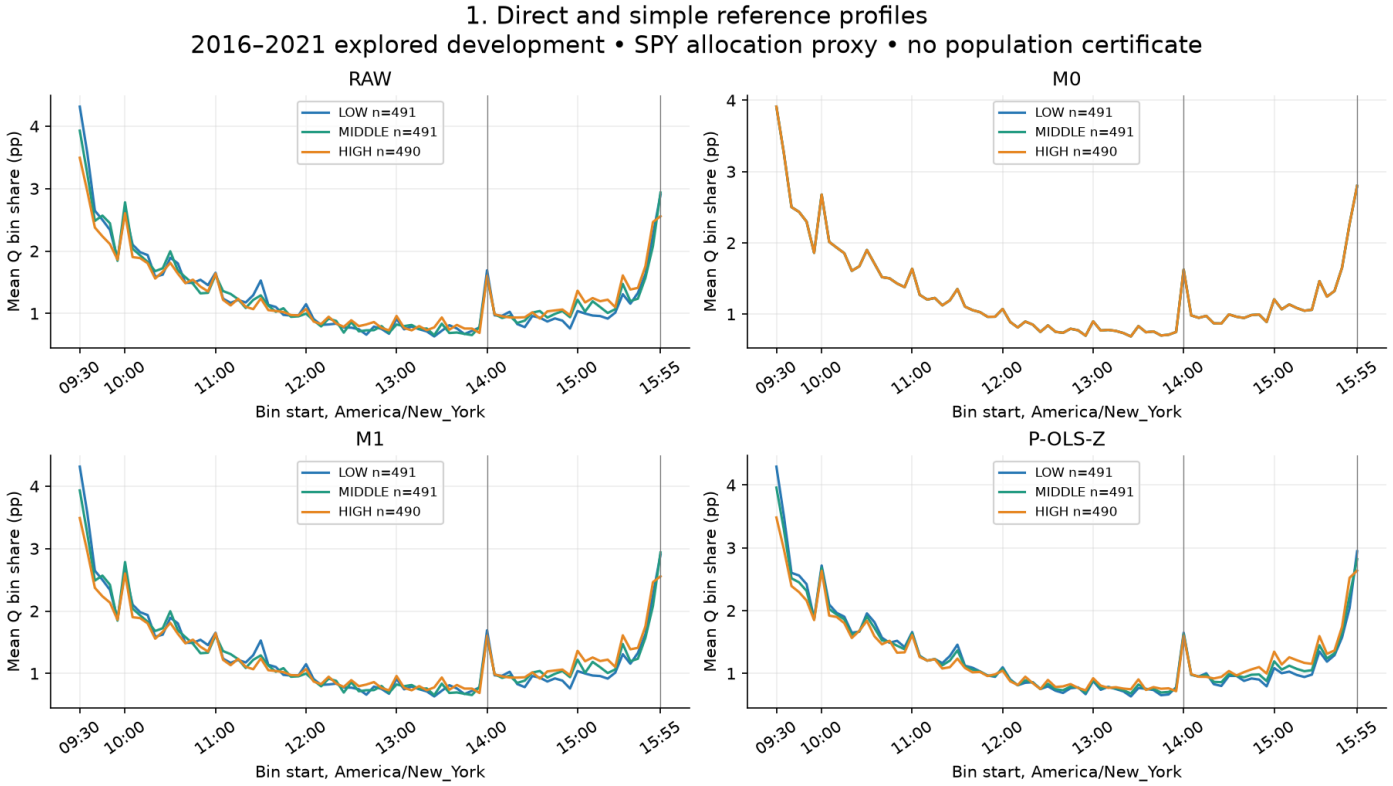
C3 scientific gallery for human review

All figures use SPY 2016–2021 previously explored development profiles. LOW is blue, MIDDLE teal, HIGH orange. Curves are complete prediction averages over the same observed state rows, except explicitly named singleton model slices. All means and contrasts use 78-bin whole-day Q. D_FULL n=1,472 is in-sample; DEV2020/2021 n=241/248 are prior-explored backchecks. No plot establishes population truth or certified materiality.

model	H1	H1_error	H3_TV	half_L1
RAW	2.00498	0	0.0485101	0.0485101
T0_EBM_TIME_ONLY	6.93889e-16	-2.00498	1.30104e-17	1.30104e-17
M3c_EBM_ADDITIVE	5.0307e-15	-2.00498	7.80626e-17	7.80626e-17
M3d_EBM_ADDITIVE	6.93889e-16	-2.00498	3.51282e-17	3.51282e-17
M5_EBM_CONT_INTERACT	1.56544	-0.439544	—	0.0338914
M6_EBM_DISC_INTERACT	1.39458	-0.610403	—	0.0298352
M8	1.90375	-0.101233	0.0431457	0.0431457

H1/H1 error columns above are pp. H3/half-L1 are fractions; H3 blank means raw closure invalid. The three state colors identify the same states across models; line style and panel title identify model and source.

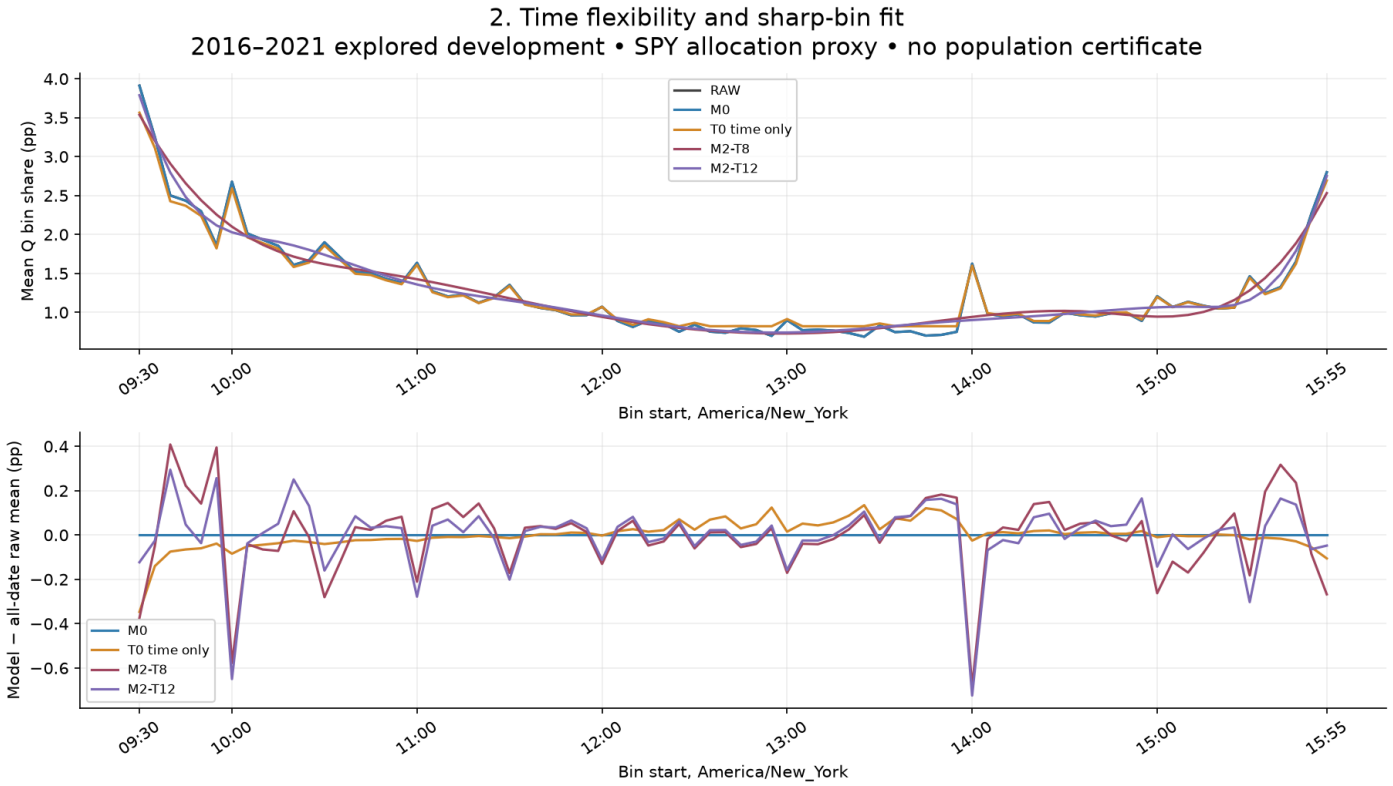
01. direct and simple reference profiles



Raw H1 +2.005pp; sharp 14:00 and 15:55 bins persist. P-OLS-Z is a strong simple shape comparator.

Numerical source: `group_mean_profiles.csv: D_FULL/HISTORICAL_P0; mean,bin,state`. All paths refer to `artifacts/C/C3_one_state_ebm_v1_1/`.

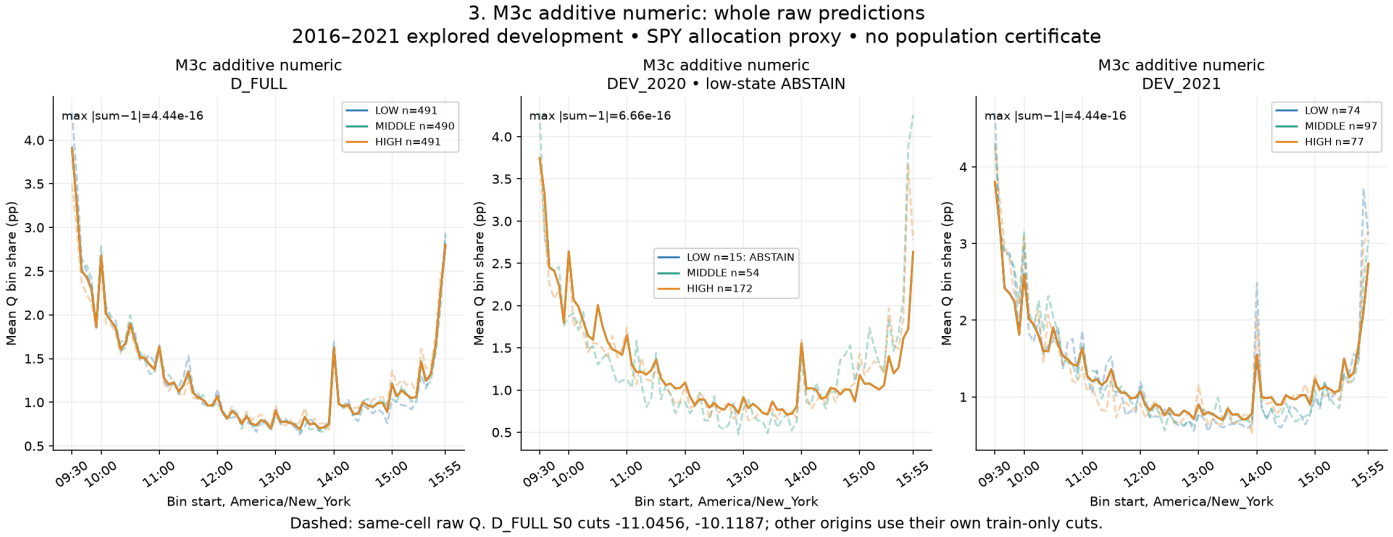
02. time only ebm vs unsmoothed and splines



T0 retains all 78 time categories. Its conservative finite-round fit still shrinks sharp time structure; M0 is the exact unsmoothed mean.

Numerical source: `global_mean_profiles.csv: means/bias`. All paths refer to `artifacts/C/C3_one_state_ebm_v1_1/`.

03. additive ebm continuous three state curves

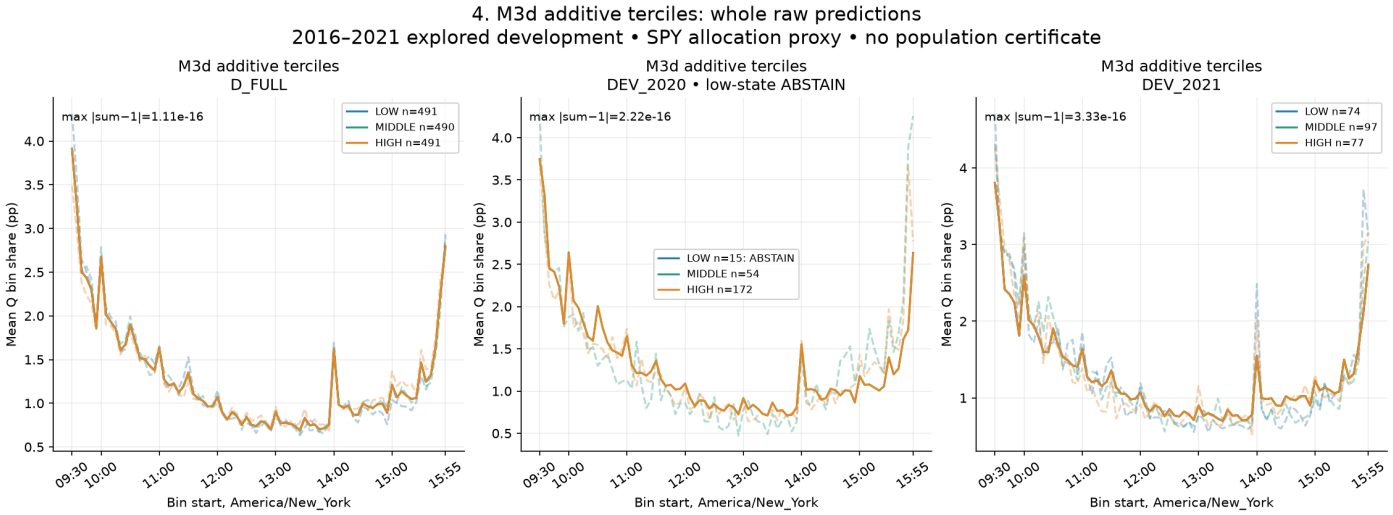


Dashed: same-cell raw Q. D_FULL S0 cuts -11.0456, -10.1187; other origins use their own train-only cuts.

Additive state effects collapse to numerical zero on balanced normalized targets; native interactions redistribute allocation but do not enforce closure. Forecast curves are fixed at the origin. 2020 low n=15 under past-origin S0 (n=21 under full-S0): ABSTAIN for raw high-low inference.

Numerical source: `group_mean_profiles.csv: TRAIN_S0; profile_predictive_metrics.csv: sum_defect_max`. All paths refer to `artifacts/C/C3_one_state_ebm_v1_1/`.

04. additive ebm discrete three state curves

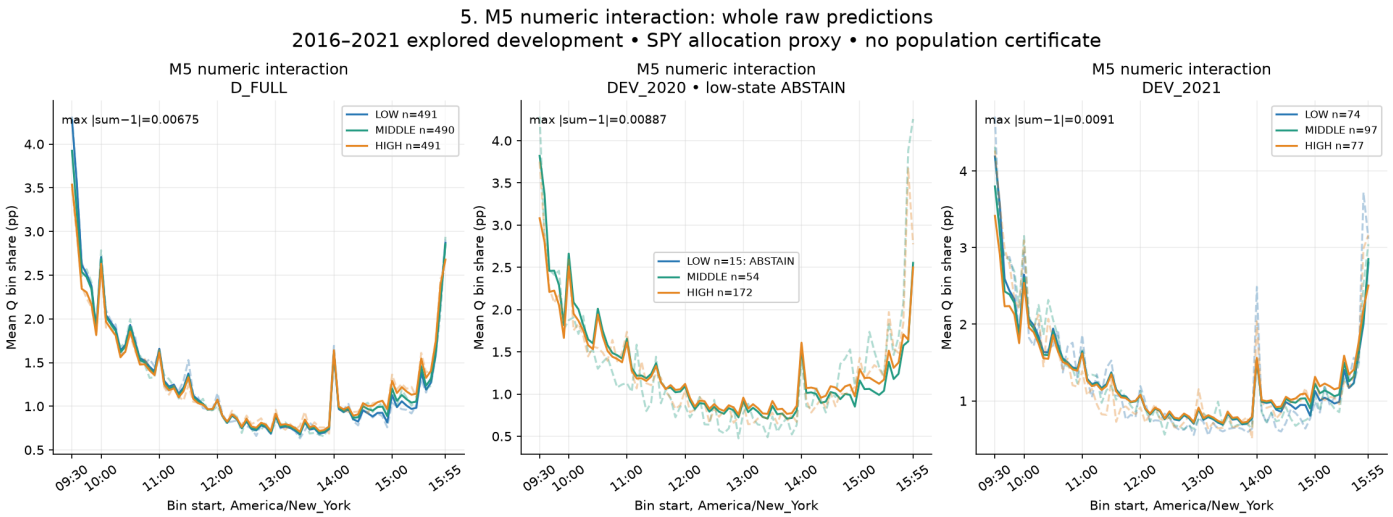


Dashed: same-cell raw Q. D_FULL S0 cuts -11.0456, -10.1187; other origins use their own train-only cuts.

Additive state effects collapse to numerical zero on balanced normalized targets; native interactions redistribute allocation but do not enforce closure. Forecast curves are fixed at the origin. 2020 low n=15 under past-origin S0 (n=21 under full-S0): ABSTAIN for raw high-low inference.

Numerical source: `group_mean_profiles.csv: TRAIN_S0; profile_predictive_metrics.csv: sum_defect_max`. All paths refer to `artifacts/C/C3_one_state_ebm_v1_1/`.

05. continuous interaction ebm three state curves

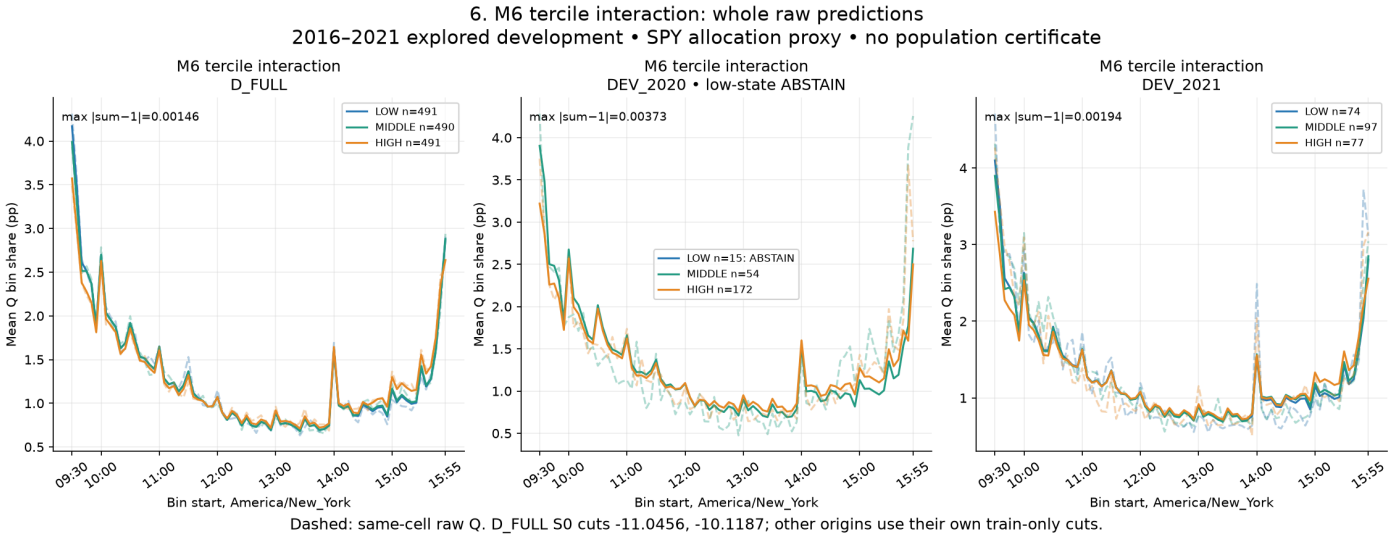


Dashed: same-cell raw Q. D_FULL S0 cuts -11.0456, -10.1187; other origins use their own train-only cuts.

Additive state effects collapse to numerical zero on balanced normalized targets; native interactions redistribute allocation but do not enforce closure. Forecast curves are fixed at the origin. 2020 low n=15 under past-origin S0 (n=21 under full-S0): ABSTAIN for raw high-low inference.

Numerical source: `group_mean_profiles.csv: TRAIN_S0; profile_predictive_metrics.csv: sum_defect_max`. All paths refer to `artifacts/C/C3_one_state_ebm_v1_1/`.

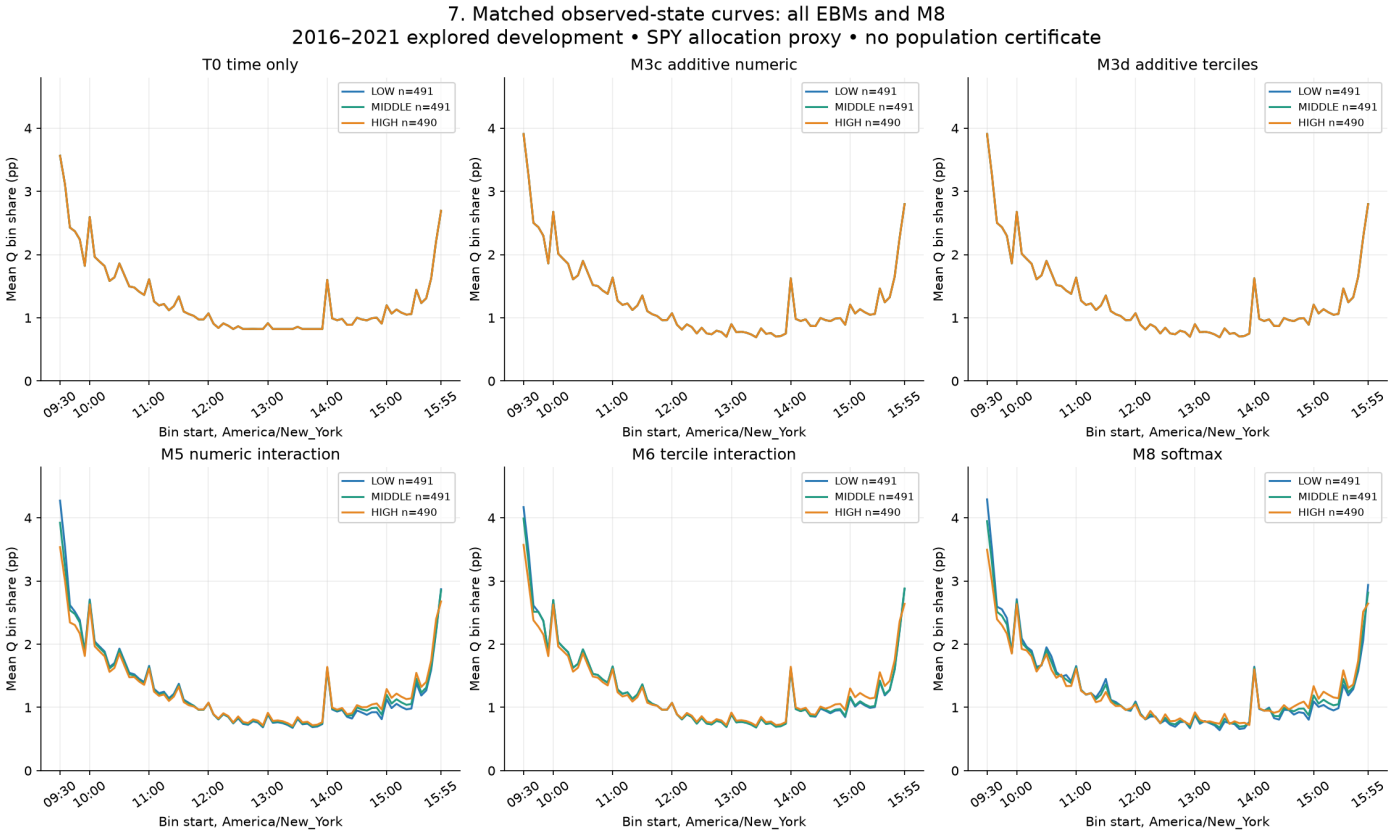
06. discrete interaction ebm three state curves



Additive state effects collapse to numerical zero on balanced normalized targets; native interactions redistribute allocation but do not enforce closure. Forecast curves are fixed at the origin. 2020 low n=15 under past-origin S0 (n=21 under full-S0): ABSTAIN for raw high-low inference.

Numerical source: `group_mean_profiles.csv: TRAIN_S0; profile_predictive_metrics.csv: sum_defect_max`. All paths refer to `artifacts/C/C3_one_state_ebm_v1_1/`.

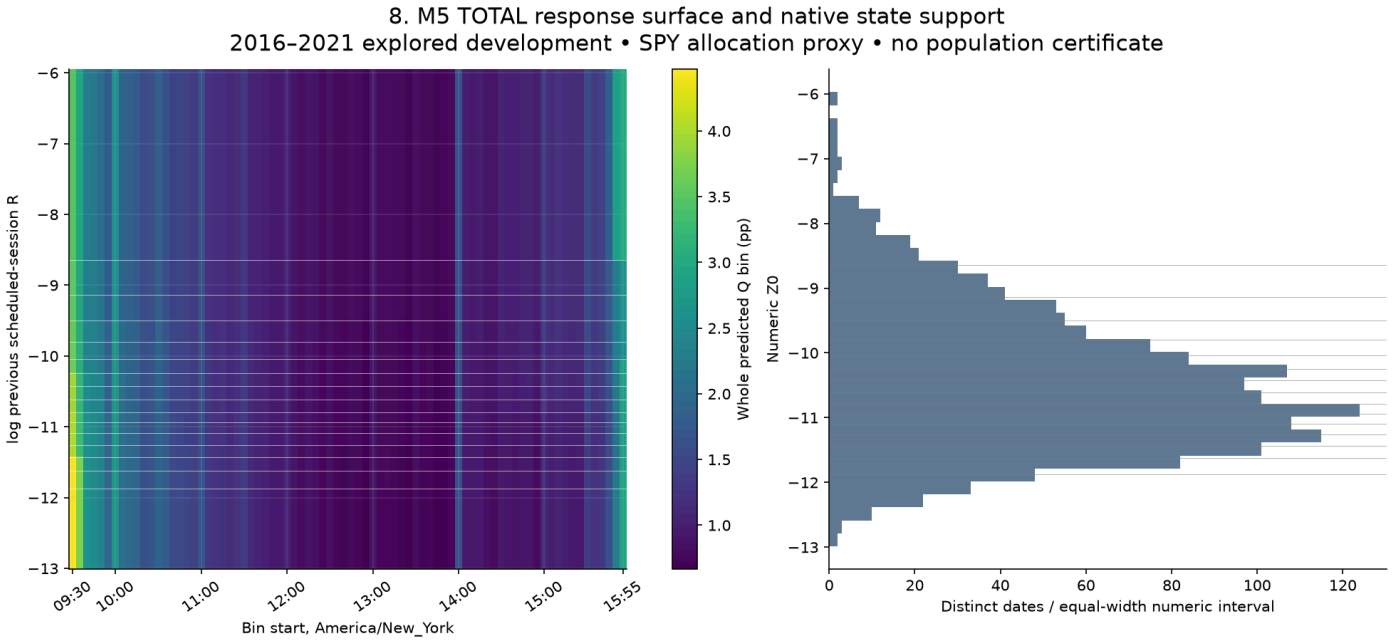
07. all ebms matched shapes overview



Same state colors and y scale. The additive fits share one shape; M5/M6 retain modest state/time structure; M8 is intrinsically normalized.

Numerical source: `group_mean_profiles.csv: D_FULL/HISTORICAL_P0; identical rows`. All paths refer to `artifacts/C/C3_one_state_ebm_v1_1/`.

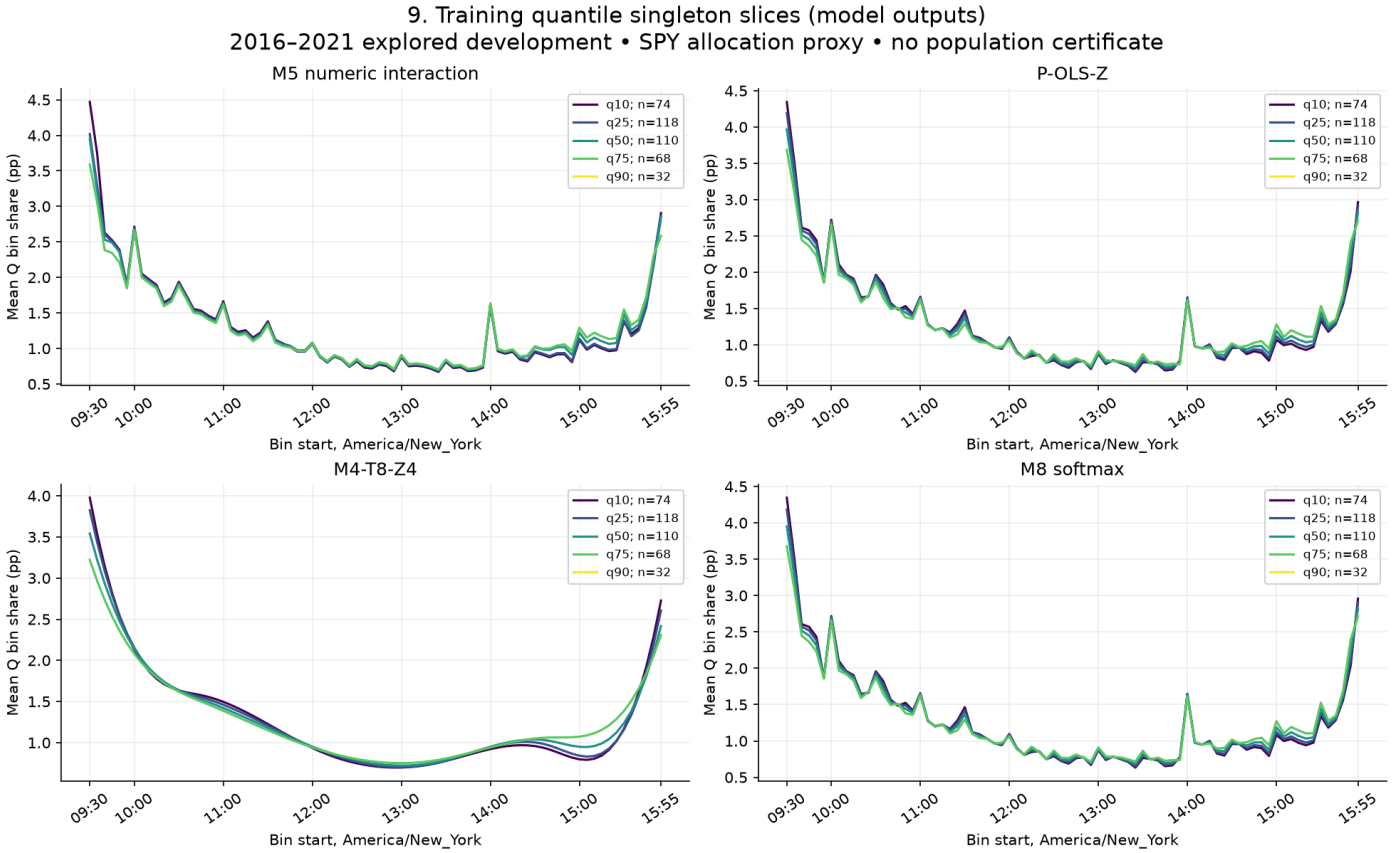
08. continuous ebm total surface and support



Stepwise numeric EBM total prediction, never an isolated interaction term. Bands failing basic date/group support are masked; all D_FULL 16 quantile bins currently pass.

Numerical source: `continuous_total_surface.csv; checkpoint D_FULL M5 transform.cuts`. All paths refer to `artifacts/C/C3_one_state_ebm_v1_1/`.

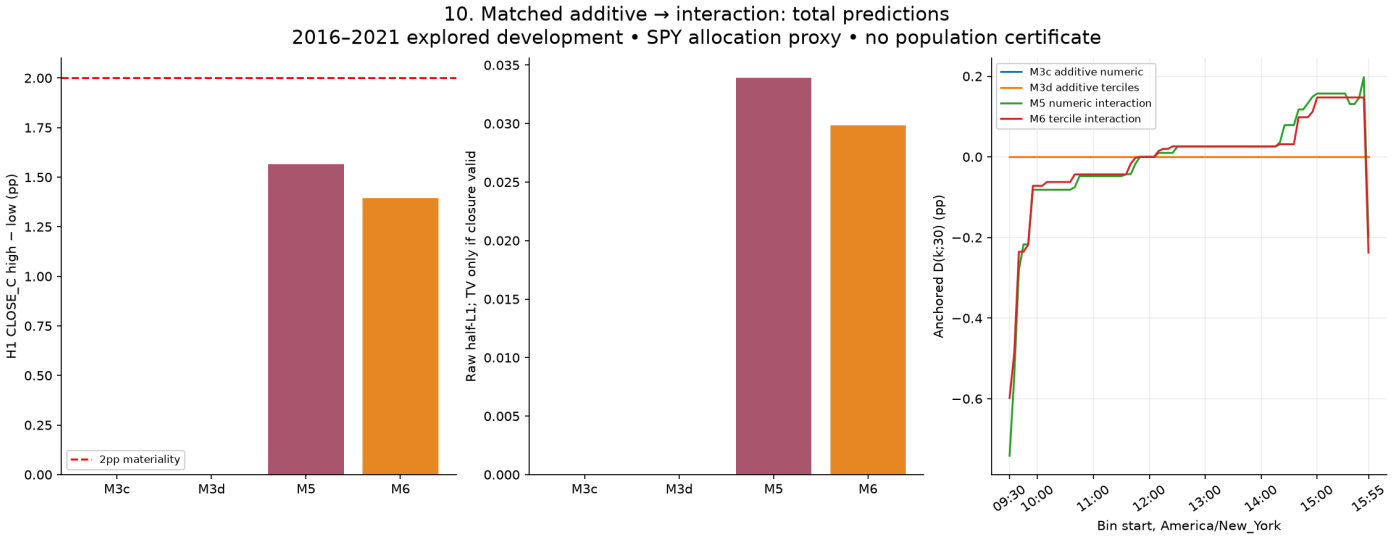
09. continuous ebm five numeric state slices



All models evaluated at identical train-only quantiles. A singleton fitted curve is not an identified population conditional mean; neighborhoods describe support.

Numerical source: `continuous_slices.csv: prediction; neighborhood width and date/group counts`. All paths refer to `artifacts/C/C3_one_state_ebm_v1_1/`.

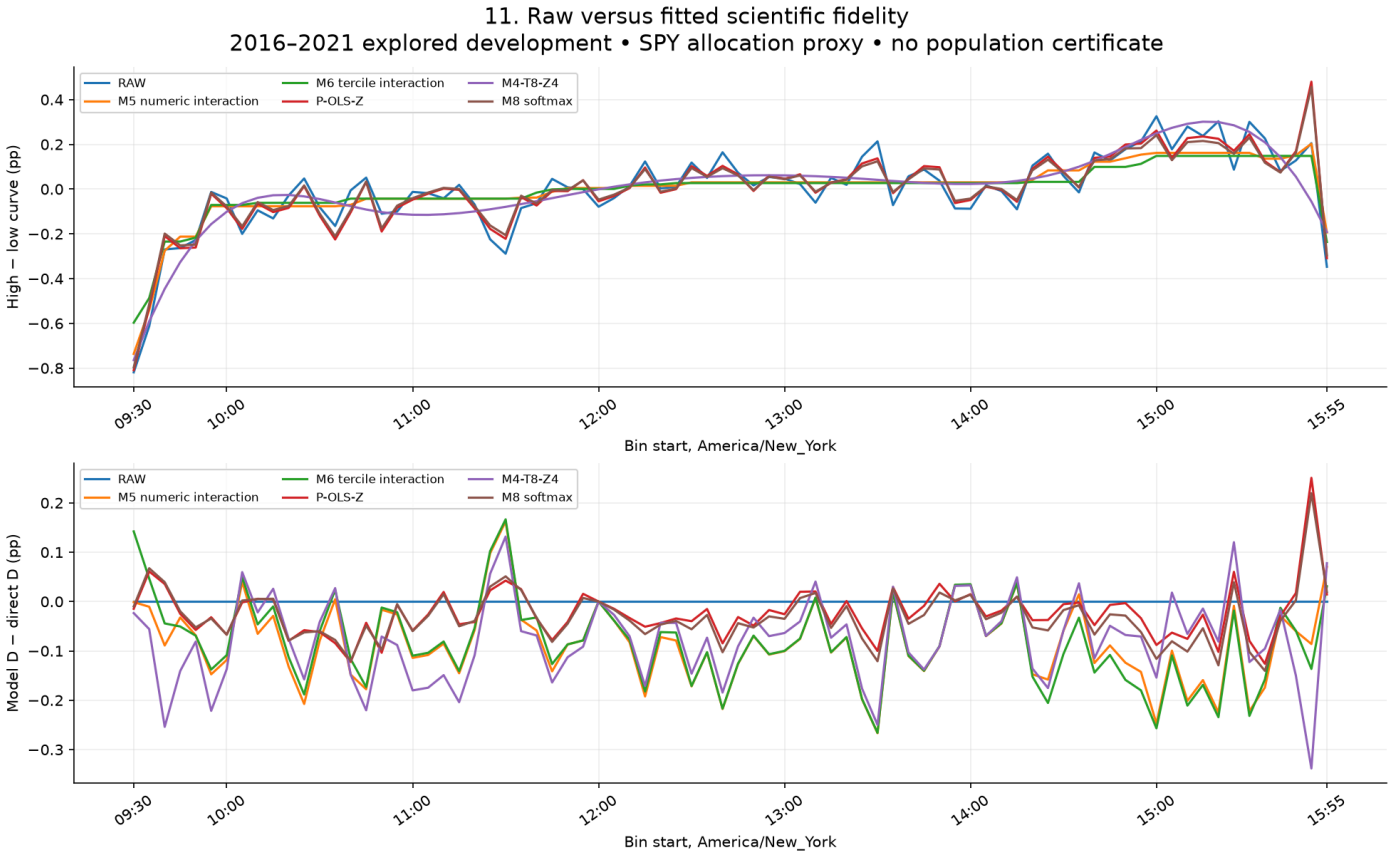
10. native additive vs interaction contrast



Raw M3 anchored D is zero to roundoff. M5/M6 whole-model contrasts are nonzero but attenuate raw H1. Invalid-closure half-L1 is explicitly not TV.

Numerical source: `model_shapes_and_scientific_contrasts.csv; group_mean_profiles.csv anchored_D`. All paths refer to `artifacts/C/C3_one_state_ebm_v1_1/`.

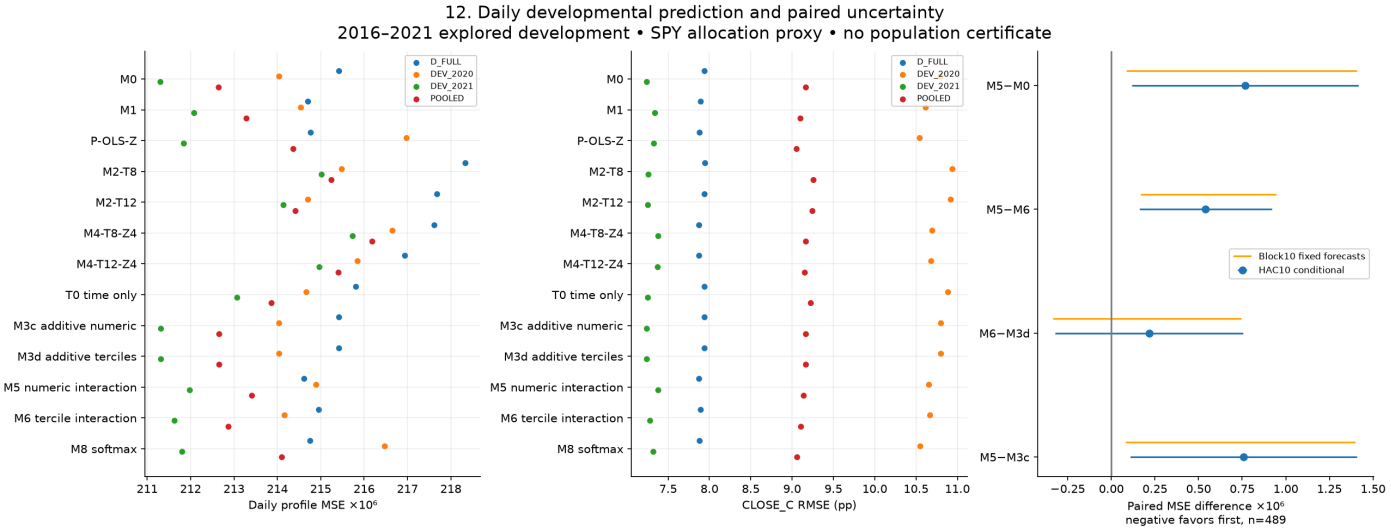
11. raw versus fitted scientific fidelity



M5 H1≈1.565pp, M6≈1.395pp versus raw≈2.005pp. P-OLS-Z and M8 are stronger H1 fidelity references; no extra EBM fidelity certificate.

Numerical source: `group_mean_profiles.csv: matched historical P0 state_difference/anchored_D/raw_anchored_D`. All paths refer to `artifacts/C/C3_one_state_ebm_v1_1/`.

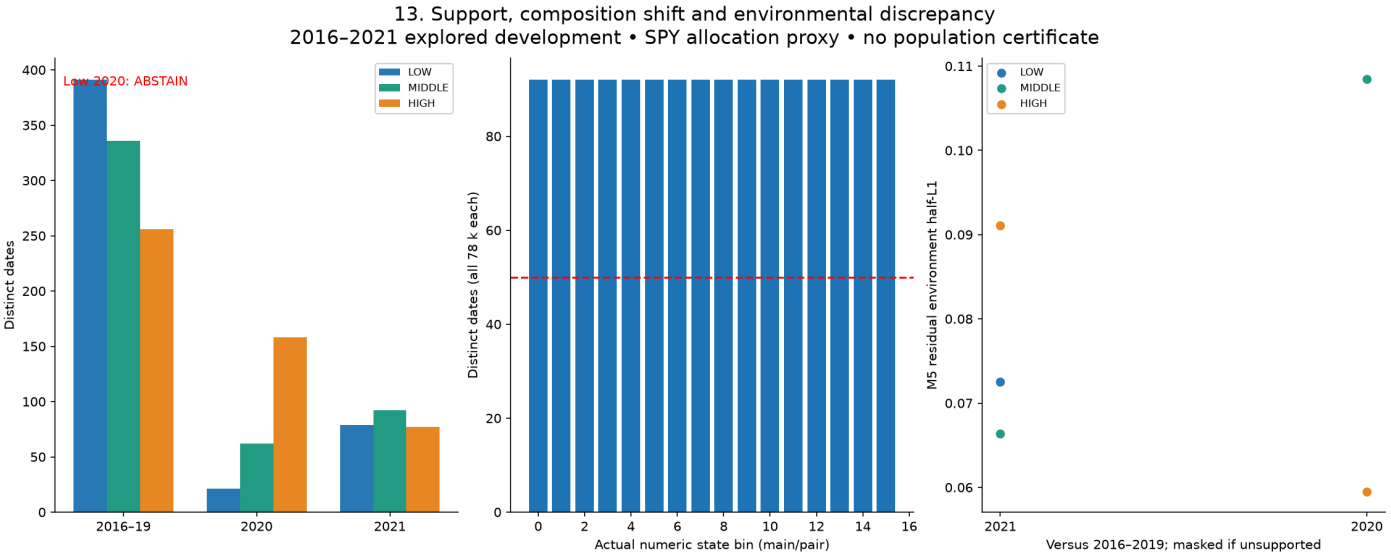
12. developmental error scorecard



2020 n=241, 2021 n=248, pooled n=489. Native interaction adds no established profile increment over matched additive/M0. Confidence labels condition on saved selected forecasts.

Numerical source: `profile_predictive_metrics.csv; paired_loss_tests.csv HAC10 and block10`. All paths refer to `artifacts/C/C3_one_state_ebm_v1_1/`.

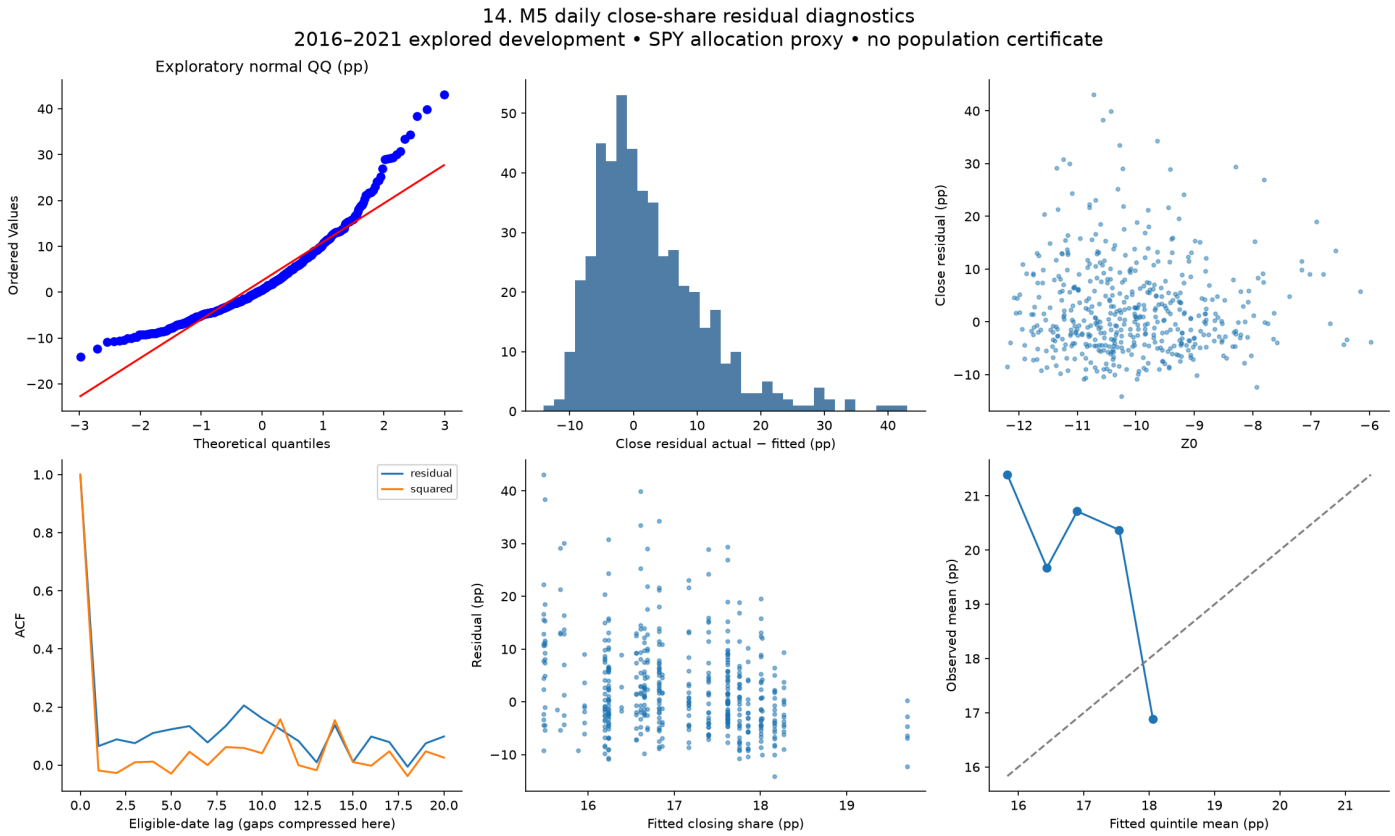
13. state support year composition and drift



Quantile occupancy over full development does not ensure yearly overlap. Basic gates ≥50 dates/20 original groups are operational; absent overlap remains ABSTAIN.

Numerical source: `state_support_and_masks.csv; environment_residual_discrepancy.csv`. All paths refer to `artifacts/C/C3_one_state_ebm_v1_1/`.

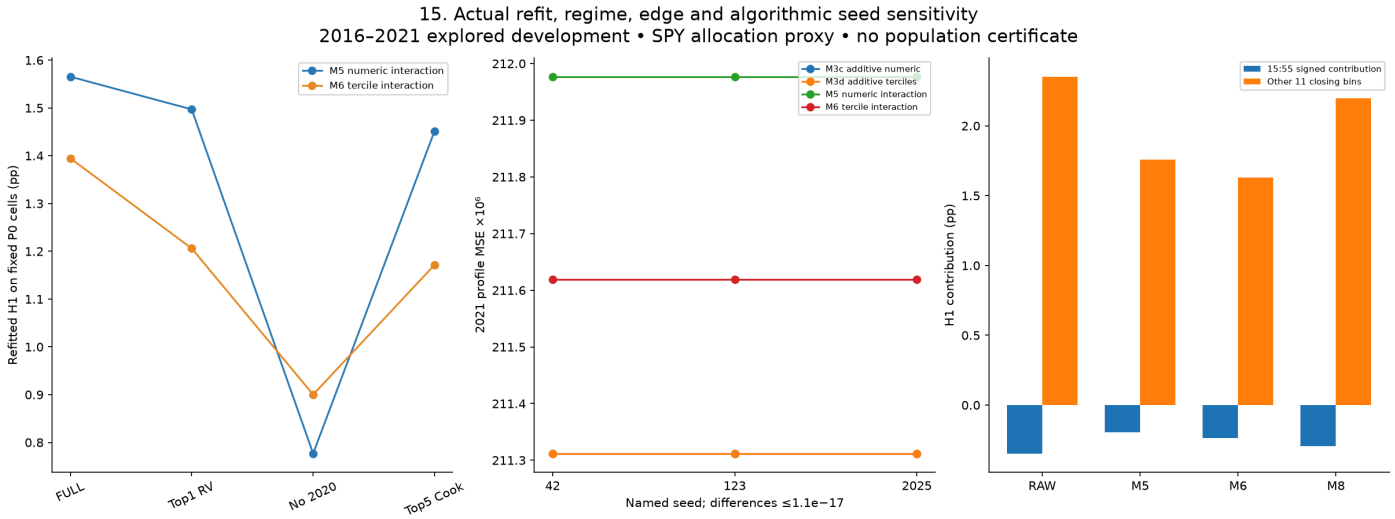
14. residual qq acf arch and calibration



Heavy tails and residual dependence persist. ARCH/Ljung–Box references are heuristic; native EBM has no classical OLS Cook/hat inference.

Numerical source: `residual_daily_features.csv: POOLED/M5; calibration_quintiles.csv; serial_diagnostics.csv`. All paths refer to `artifacts/C/C3_one_state_ebm_v1_1/`.

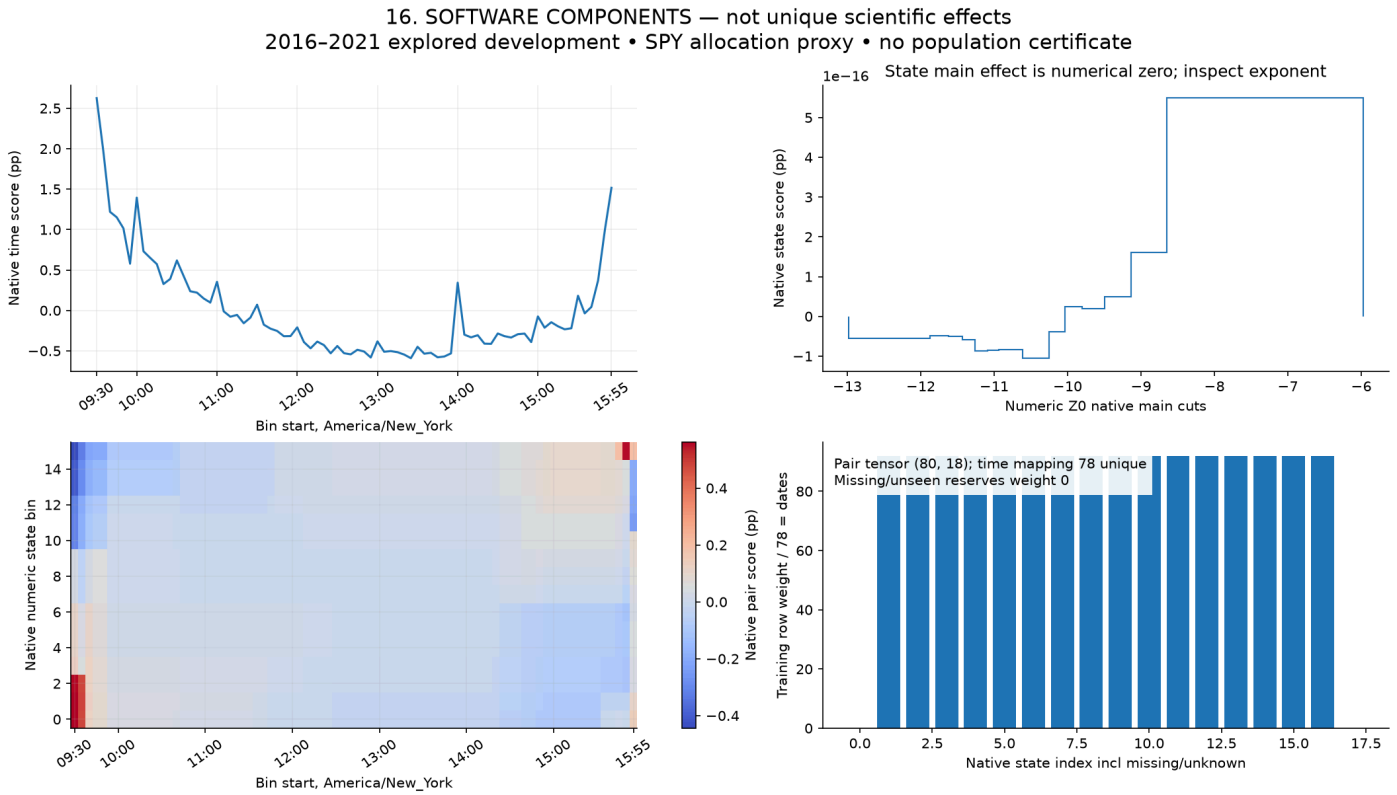
15. outlier regime seed stability



Ten actual stress refits completed; top5 additive controls and native LOYO deferred. All three seeds are deterministic to roundoff. Last-bin exclusion is exploratory, primary CLOSE_C unchanged.

Numerical source: `stress_refits_and_fixed_scores.csv; native_seed_and_complexity_stability.csv; model_shapes_and_scientific_contrasts.csv`. All paths refer to `artifacts/C/C3_one_state_ebm_v1_1/`.

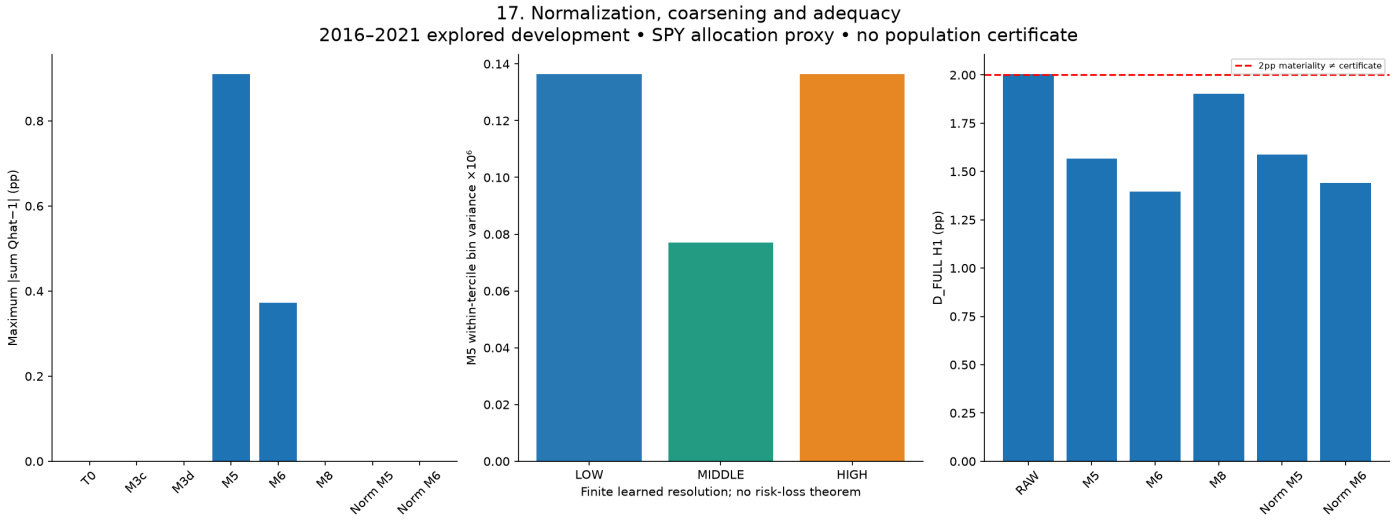
16. native software terms and binning



Main and pair components depend on centering and software conventions. Scientific contrasts elsewhere use total native predict. Shared category mappings may be stored once and reused at both resolutions.

Numerical source: `native_terms_and_bins.json + hash-verified D_FULL M5 checkpoint: term_scores_,bins_,bin_weights_`. All paths refer to `artifacts/C/C3_one_state_ebm_v1_1/`.

17. normalization coarsening model adequacy



NORM variants apply the frozen positive-part closure map and have distinct model identities. They cannot inherit native additive interpretation. No model supplies a raw materiality/equivalence certificate.

Numerical source: `profile_predictive_metrics.csv; continuous_vs_discrete_coarsening.csv; model_shapes_and_scientific_contrasts.csv`. All paths refer to `artifacts/C/C3_one_state_ebm_v1_1/`.

Figure/source provenance register

figure	numerical_source	sha256
01_direct_and_simple_reference_profiles.png	group_mean_profiles.csv: D_FULL/HISTORICAL_P0; mean,bin,state	d14081dabd67fcedeb8f440711658f3e3b4305b7b65
02_time_only_ebm_vs_unsmoothed_and_splines.png	global_mean_profiles.csv: means/bias	11b9413129a5f5e803588a37fa25705b8234019664
03_additive_ebm_continuous_three_state_curves.png	group_mean_profiles.csv: TRAIN_S0; profile_predictive_metrics.csv: sum_defect_max	8d07ee11069bea8990b60e712777e3f628101969a3

figure	numerical_source	sha256
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05_continuous_interaction_ebm_three_state_curves.png	group_mean_profiles.csv: TRAIN_S0; profile_predictive_metrics.csv: sum_defect_max	8da58d67f3f1d76a2880827d231dcec27b21996af46
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07_all_ebms_matched_shapes_overview.png	group_mean_profiles.csv: D_FULL/HISTORICAL_P0; identical rows	ebd1c4dbec91ad1d3a4f004ee958e4063823fa8af92
08_continuous_ebm_total_surface_and_support.png	continuous_total_surface.csv; checkpoint D_FULL M5 transform.cuts	2e60279a8a2684c03a2745f1a6753d04e871905e6a
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10_native_additive_vs_interaction_contrast.png	model_shapes_and_scientific_contrasts.csv; group_mean_profiles.csv anchored_D	5dd25d4c2a0c7adbeaf8fdd163ebba6d4ba4ce74ef0
11_raw_versus_fitted_scientific_fidelity.png	group_mean_profiles.csv: matched historical P0 state_difference/anchored_D/raw_anchored_D	6902960319d699135b1d6d9fde56da440d9db360a
12_developmental_error_scorecard.png	profile_predictive_metrics.csv; paired_loss_tests.csv HAC10 and block10	da873307fc2b27cc2a11d38a4b1a18b385d14d5f5a90
13_state_support_year_composition_and_drift.png	state_support_and_masks.csv; environment_residual_discrepancy.csv	1e118847c14cf0983c739218e89ad07b37d8937a620
14_residual_qq_acf_arch_and_calibration.png	residual_daily_features.csv: POOLED/M5; calibration_quintiles.csv; serial_diagnostics.csv	2b0205e9b08879de82ce21c555749902ed5fb9e88
15_outlier_regime_seed_stability.png	stress_refits_and_fixed_scores.csv; native_seed_and_complexity_stability.csv; model_shapes_and_scientific_contrasts.csv	704ad0316b8659067bf2038c95d1353cbcf8041bb3
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